MP307 : Modelling II

Cian Mac Liatháin
11476078

Q. 1 (a) (i) $f(t) = \beta e^{-\beta t}, \quad t \geq 0$

$$\mu = \langle t \rangle = \beta \int_0^\infty t e^{-\beta t} dt$$

$$= \frac{1}{\beta}$$

$$\int u\,dv = uv - \int v\,du$$

$u = t \qquad dv = e^{-\beta t}$

$du = dt \qquad v = \frac{-e^{-\beta t}}{\beta}$

$$= \frac{-te^{-\beta t}}{\beta} - \int \frac{-e^{-\beta t}}{\beta} dt$$

$$= \frac{-te^{-\beta t}}{\beta} - \frac{e^{-\beta t}}{\beta^2} \Big|_0^\infty$$

$$= \frac{1}{\beta^2}$$

(ii) Variance $= \langle (t-\mu)^2 \rangle = \langle t^2 \rangle - \mu^2$

$$\sigma^2 = \int_0^\infty t^2 \beta e^{-\beta t} dt - \frac{1}{\beta^2}$$

$$= \frac{2}{\beta^2} - \frac{1}{\beta^2}$$

$$= \frac{1}{\beta^2}$$

$$\sigma = \sqrt{1/\beta^2}$$

(iii) $\beta = 1/5 \,/\text{min}$, $\mu = \frac{1}{\beta} = \frac{1}{1/5} = 5$ min on average for customer to be served

$\therefore \frac{60}{5} = 12$ customers served per hour

(b) (i) $\sum_{j=0}^{r} p(i,j) = 1$:

$p + 0.2 + 0 = 1$ $\quad$ $p = 0.8$

$0 + q + 0.3 = 1$ $\quad$ $q = 0.7$

(ii) $P = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0 & 0.7 & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix}$

$P^2 = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0 & 0.7 & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.64 & 0.2 & 0.06 \\ 0.15 & 0.49 & 0.36 \\ 0.65 & 0.1 & 0.25 \end{pmatrix}$

$p(0,1)$ in two time-steps is 0.2

(iii) A Markov process is said to be ergodic

A Markov chain is ergodic if:

$\pi_j \equiv \lim_{m \to \infty} p_m(i,j)$

This Markov process can be shown to be ergodic if all the states of the system may be visited from any other state in a finite number m transitions.

We can see this is true when $m = 2$ because all entries of $P^2 \neq 0$

(iv) A row vector of equilibrium probabilities $\pi = (\pi_0, \pi_1, \ldots, \pi_r)$ satisfies:

$\pi = \pi P$

$\therefore$ π is a left eigenvector of P with a unique eigenvalue of 1

$\pi = \pi P$: $(\pi_0, \pi_1, \pi_2) = (\pi_0, \pi_1, \pi_2) \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0 & 0.7 & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix}$

$\pi_2 = 1 - \pi_0 - \pi_1$

$$(\pi_0, \pi_1, 1-\pi_0-\pi_1) = (\pi_0, \pi_1, 1-\pi_0-\pi_1)\begin{pmatrix} 0.8 & 0.2 & 0 \\ 0 & 0.7 & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix}$$

$\pi_0 = 0.8\pi_0 + (1-\pi_0-\pi_1)0.5$

$\pi_0 = 0.8\pi_0 - 0.5\pi_0 - 0.5\pi_1 + 0.5$

$0.7\pi_0 = 0.5 - 0.5\pi_1$

$\frac{7}{10}\pi_0 = \frac{1}{2} - \frac{1}{2}\left(\frac{2}{3}\right)\pi_0$

$\frac{7}{10}\pi_0 + \frac{2}{6}\pi_0 = \frac{1}{2}$

$\frac{31}{30}\pi_0 = \frac{1}{2}$

$\pi_0 = \frac{15}{31} = 0.4839$

$\pi_1 = 0.2\pi_0 + 0.7\pi_1$

$0.3\pi_1 = 0.2\pi_0$

$\pi_1 = \frac{2}{3}\pi_0$

$\pi_1 = \frac{2}{3}\left(\frac{15}{31}\right)$

$\pi_1 = \frac{10}{31} = 0.3226$

$\pi_2 = 1-\pi_0-\pi_1$

$\pi_2 = 1 - \frac{15}{31} - \frac{10}{31}$

$\pi_2 = \frac{6}{31} = 0.1935$

$$\pi = \left\{\frac{15}{31}, \frac{10}{31}, \frac{6}{31}\right\}$$

(c) $\frac{dS}{dt} = -\alpha SI$, $\frac{dR}{dt} = \beta I$ $\alpha, \beta > 0$ and p

then $\frac{dS}{dR} = \frac{dS}{dt} \cdot \frac{dt}{dR} = \frac{-\alpha SI}{\beta I} = \frac{-\alpha S}{\beta}$

$\int \frac{1}{S}\, dS = \int \frac{-\alpha}{\beta}\, dR$

$\ln(S) + S_0 = \frac{-\alpha}{\beta}R$

$S(R) = e^{\frac{-\alpha R}{\beta}}$

(ii) $\frac{dI}{dt} = \frac{-dS}{dt} - \frac{dR}{dt} = (\alpha S - \beta)I$ if $S < \beta/\alpha$ then $\frac{dI}{dt} < 0$

then the infected group die out without spreading through the general population d)

$$\frac{dx_1}{dt} = x_1 - 2x_2 + u$$

$$\frac{dx_2}{dt} = Kx_1$$

$$y = x_1$$

$$\frac{dx}{dt} = \begin{pmatrix} 1 & -2 \\ K & 0 \end{pmatrix} x + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u \quad , \quad y = \begin{pmatrix} 1 & 0 \end{pmatrix} x$$

(1)
(ii) System is Reachable if and only if its reachability matrise

$$W_r = \begin{pmatrix} B & AB & \dots & A^{n-1}B \end{pmatrix}$$

is invertible.

$$W_r = \begin{pmatrix} B & AB \end{pmatrix} \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$W_r = \begin{pmatrix} 1 & 1 \\ 0 & K \end{pmatrix} \qquad AB = \begin{pmatrix} 1 & -2 \\ K & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ K \end{pmatrix}$$

$$\det(W_r) = K$$

$\therefore$ W_r is invertible

and system is reachable if $K \neq 0$

(2) System is observable if and only if its observability matrise

$$W_o = \begin{pmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{pmatrix}$$

is of full rank

$$W_o = \begin{pmatrix} C \\ CA \end{pmatrix} \qquad C = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$CA = \begin{pmatrix} 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ K & 0 \end{pmatrix} = \begin{pmatrix} 1 & -2 \end{pmatrix}$$

$$W_o = \begin{pmatrix} 1 & 0 \\ 1 & -2 \end{pmatrix}$$

Rows are linearly independent

$\therefore$ W_r is of full rank $\therefore$ system is observable

2 (a) (i) $p_K(t)$ = Prob(K customers queuing at time t) where $K = 0, 1, 2, 3 \ldots$

After one time interval Δt:

$$p_0(t+\Delta t) = p_0(t)(1-\alpha_0\Delta t) + p_1(t)\beta_1\Delta t$$

$$p_K(t+\Delta t) = p_{K-1}(t)\alpha_{K-1}\Delta t + p_K(t)(1-(\alpha_K+\beta_K))\Delta t + p_{K+1}(t)\beta_{K+1}\Delta t \qquad \text{for } 0<K<r$$

Then $$\frac{dp_0}{dt} = \lim_{\Delta t\to 0}\frac{p_0(t+\Delta t)-p_0(t)}{\Delta t} = \lim_{\Delta t\to 0}\frac{p_0(t)(-\alpha\Delta t)+p_1(t)\beta_1\Delta t}{\Delta t} = -\alpha_0 p_0 + \beta_1 p_1$$

$$\frac{dp_K}{dt} = \lim_{\Delta t\to 0}\frac{p_K(t+\Delta t)-p_K(t)}{\Delta t} = \lim_{\Delta t\to 0}\frac{p_{K-1}(t)\alpha_{K-1}\Delta t - p_K(t)(\alpha_K+\beta_K)\Delta t + p_{K+1}(t)\beta_{K+1}\Delta t}{\Delta t} = \alpha_{K-1}p_{K-1} - (\alpha_K+\beta_K)p_K + \beta_{K+1}p_{K+1}$$

$$\frac{dp_0}{dt} = -\alpha p_0 + \beta(1-p_0)$$

$$\frac{dp_0}{dt} + (\alpha+\beta)p_0 = \beta$$

$$\frac{d}{dt}\left(e^{(\alpha+\beta)t}p_0\right) = \beta e^{(\alpha+\beta)t}$$

$$p_0(t) = \frac{\beta}{\alpha+\beta} + Ae^{-(\alpha+\beta)t} \quad \text{where } A = p_0(0)$$

$$\pi_0 = \lim_{t\to\infty} p_0(t) = \frac{\beta}{\alpha+\beta}$$

$$\pi_1 = \lim_{t\to\infty} p_1(t) = \frac{\alpha}{\alpha+\beta}$$

(ii) As the queue is unable to grow bigger than max size r, the probability of an arrival α_r must be 0

When the queue is at size 0, their probability of a serving β_0 must be 0 as there are no customers to serve.

$$\therefore \alpha_r = \beta_0 = 0$$

(iii) Since the Markov process is ergodic

equilibrium probabilities are given by $\pi_K = \lim_{t\to\infty} P_K(t)$ exists so as $t \to \infty$ $P_K(t) \to \pi_K$

$\frac{dp_K}{dt} \to 0$

from (i)

$$(i)\ 0 = -\alpha_0\pi_0 + \beta_1\pi_1$$

$$(ii)\ 0 = \alpha_{K-1}\pi_{K-1} - (\alpha_K+\beta_K)\pi_K + \beta_{K+1}\pi_{K+1} \qquad 0<K<r$$

Proof by induction

$K=1$ $\quad (i)\ \pi_1 = \frac{\alpha_0\pi_0}{\beta_1} = \rho_1\pi_0$

assume $K=K$ $\quad \pi_K = \rho_K\pi_{K-1} = \frac{\alpha_{K-1}\pi_K}{\beta_K}$

$$\beta_K\pi_K = \alpha_{K-1}\pi_{K-1}$$

Then

$$(ii)\quad 0 = \beta_K\pi_K - (\alpha_K+\beta_K)\pi_K + \beta_{K+1}\pi_{K+1}$$

$K+1$ $\quad \beta_{K+1}\pi_{K+1} = \alpha_K\pi_K$

$\pi_{K+1} = \rho_{K+1}\pi_K$ $\qquad$ true for $K = 1, 2, \dots, r$

So then $\quad \pi_K = \rho_K\rho_{K-1}\cdots\rho_1\pi_0$

(b) $\alpha_K = \alpha \quad , \beta_K = K\beta \quad , \rho = \frac{\alpha}{\beta}$

$\pi_K = \rho_K\rho_{K-1}\cdots\rho_1\pi_0$ $\qquad \rho_K = \frac{\alpha}{K\beta} = \frac{1}{K}\rho$

$\sum_{K=0}^{r}\pi_K = 1$ $\qquad \rho_K\rho_{K-1}\cdots\rho_1 = \frac{\rho^K}{K!}$

$$\sum_{K=0}^{r}\rho_K\rho_{K-1}\cdots\rho_1\pi_0 = 1$$

$$\pi_0 = \frac{1}{\sum_{K=0}^{r}\frac{\rho^K}{K!}}$$

Then $\quad \pi_K = \frac{\rho^K}{K!}\left(\frac{1}{\sum_{K=0}^{r}\frac{\rho^K}{K!}}\right)$

$$= \frac{\rho^K}{K!}e^{-\rho} \qquad K = 0, 1, 2 \dots$$

(c) $\alpha = 200$ arrivals/min

$\beta = 5$ servings/sec $= 300$/min

$\rho = \frac{200}{300} = \frac{2}{3}$

$P(k>2) = 1 - P(k=0) - P(k=1) - P(k=2)$

$\pi_0 = \frac{\left(\frac{2}{3}\right)^0}{0!} e^{-2/3} = e^{-2/3} = 0.5134$

$\pi_1 = \frac{(2/3)^1}{1!} e^{-2/3} = \frac{2}{3e^{2/3}} = 0.3422$

$\pi_2 = \frac{(2/3)^2}{2!} e^{-2/3} = \frac{4}{18e^{2/3}} = 0.2282$

$P(k>2) = 1 - 0.5134 - 0.3422 - 0.2282$
$= 0.0302$

4 $\quad \frac{dc}{dt} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} c + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u \;,\quad y = \begin{pmatrix} 0 & 1 \end{pmatrix} c \qquad c = \begin{matrix} c_1 \\ c_2 \end{matrix}$

(a) $W_r = \begin{pmatrix} B & AB \end{pmatrix} \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad AB = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$W_r = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \qquad\qquad = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$\det(W_r) = 1$ $\quad\therefore$ W_r is invertible

$\therefore$ system is reachable

$W_o = \begin{pmatrix} C \\ CA \end{pmatrix} \qquad C = \begin{pmatrix} 0 & 1 \end{pmatrix} \qquad CA = \begin{pmatrix} 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \end{pmatrix}$

$W_o = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$ Rows are linearly independent

$\therefore$ W_o is of full rank

$\therefore$ System is observable

(b) $u = -\Sigma c + pr \qquad \Sigma = (\sigma_1 \; \sigma_2) \;,\; y = r$

$= -\sigma_1 c_1 - \sigma_2 c_2 + pr \qquad \lambda_1 = \lambda_2 = -1$

$\frac{dc}{dt} = Ac + B(-\Sigma c + pr)$

$= (A - B\Sigma)c + Bpr$

Equilibrium point:

$c_e = -(A - B\Sigma)^{-1} Bpr$

$B\Sigma = \begin{pmatrix} 1 \\ 0 \end{pmatrix} (\sigma_1 \; \sigma_2) = \begin{pmatrix} \sigma_1 & \sigma_2 \\ 0 & 0 \end{pmatrix}$

$A - B\Sigma = \begin{pmatrix} 2-\sigma_1 & 1-\sigma_2 \\ 1 & 2 \end{pmatrix}$

$\frac{dc}{dt} = \begin{pmatrix} 2-\sigma_1 & 1-\sigma_2 \\ 1 & 2 \end{pmatrix} c + \begin{pmatrix} 1 \\ 0 \end{pmatrix} pr$

$\det \begin{pmatrix} 2-\sigma_1-\lambda & 1-\sigma_2 \\ 1 & 2-\lambda \end{pmatrix} = \begin{vmatrix} 2-\sigma_1-\lambda & 1-\sigma_2 \\ 1 & 2-\lambda \end{vmatrix} = (2-\sigma_1-\lambda)(2-\lambda) - (1-\sigma_2)(1)$

$= 4 - 2\lambda - 2\sigma_1 + \sigma_1\lambda - 2\lambda + \lambda^2 - 1 + \sigma_2$

$= \lambda^2 + \lambda(-4 + \sigma_1) - 2\sigma_1 + \sigma_2 + 3$

$$\lambda_1 = \lambda_2 = -1$$

$$(\lambda+1)(\lambda+1) = \lambda^2 + 2\lambda + 1 = \lambda^2 + \lambda(-4+\sigma_1) - 2\sigma_1 + \sigma_2 + 3$$

$$-4+\sigma_1 = 2 \qquad -2\sigma_1 + \sigma_2 + 3 = 1$$

$$\sigma_1 = 6 \qquad \sigma_2 = 1 - 3 + 2\sigma_1$$

$$\sigma_2 = -2 + 12$$

$$\sigma_2 = 10$$

Steady-state output:

$$y_e = Cc_e = -C(A - B\Sigma)^{-1} B p r$$

for $y_e = r$

$$r = \frac{Bpr}{-C(A-B\Sigma)}$$

$$p = -\frac{1}{C(A-B\Sigma)^{-1}B}$$

$$C(A-B\Sigma)^{-1}B = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 2-\sigma_1 & 1-\sigma_2 \\ 1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} -4 & -9 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 9 \\ -1 & -4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & -4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$-1$$

$$p = \frac{-1}{-1} = 1$$

$$u = -\sigma_1 c_1 - \sigma_2 c_2 + pr$$

$$= -6c_1 - 10c_2 + r$$